

Research papers

Relating soil-root hydraulic resistance variation to stomatal regulation in soil-plant water transport modeling

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ABSTRACT

Soil-root hydraulic resistance variation and stomatal regulation are two critical hydrophysiological responses of plants to drought stress; however, few studies have been developed to quantify their interactions. To fill this gap, we developed a soil-plant hydraulic model (SR-HRV) that attempts to characterize the effects of stomatal regulation and three universal soil-root hydraulic resistance variations, i.e., root aquaporins promotion (AQU), apoplastic path damage (APD), and root-soil contact loosening (CONTACT). The sensitive parameters of the SR-HRV model were analyzed and optimized based on a field experiment with sunflower plants (*Helianthus annuus* L.). Several simulation scenarios were designed to clarify the individual and interactive effects of soil-root hydraulic resistance variations for plants with different stomatal sensitivities. Results show that the sensitivity of simulated stomatal conductance and soil water content response to stomatal regulation parameters, especially to abscisic acid-related parameters, are more active than to soil-root hydraulic resistance variation parameters. But as the soil dries, the sensitivities to APD and CONTACT parameters are rapidly increased. The simulation demonstrates that AQP alleviates the leaf water potential drop-down and maintains relatively high root water absorption of the plant when it is in mild drought conditions, while CONTACT and APD respectively restrict the water flux and drought signal responses with continuous soil dehydration. Moreover, the AQP effects are more pronounced but the effects of APD and CONTACT would be restricted for plants with higher stomatal sensitivity to drought signals. These simulation results imply the diverse response strategies of plants to drought, the collaborations between stomatal regulation and soil-root hydraulic resistance variations should be considered in soil-plant water transport modeling.

1. Introduction

As a primary physiological process accompanying photosynthesis, plant transpiration plays an essential role in the terrestrial hydrological cycle (Evaristo et al., 2015). Precisely quantifying plant transpiration is a significant prerequisite for a number of applications including climate prediction, water resource management, and crop yield estimation (Lian et al., 2018; Wang et al., 2018). However, owing to the difficulty of estimating stomatal conductance and other complex plant hydrophysiological responses to environmental stresses, transpiration estimation suffers from large uncertainty, the range of predicted

transpiration is spanning 20–65 % of terrestrial evapotranspiration (Jasechko et al., 2013; Vereecken et al., 2015).

The radial hydraulic resistances from the rhizosphere (the soils surrounding the root surface) to the root xylem are known as the primary constraints on water transport during a drought event (Carminati and Javaux, 2020). Specifically, besides being restricted by the hydraulic status in the bulk soil, the radial water flow is also affected by the hydraulic conductivity at the soil-root interface and in root tissue. Effective root-soil contact is a premise for the continuity of root water absorption in the rhizosphere, but with soil dehydration, air gaps would emerge at the soil-root interface, which restricts the water movement from soil to root. Advances in noninvasive imaging techniques in soil science and

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Nomenclature

Acronym

ABA	Abscisic acid
LAI	Leaf area index
LAI _{eff}	Effective leaf area index
A + A + C	A simulation scenario considering three soil-root hydraulic resistance variations (AQP + APD + CONTACT)
AQP	Aquaporins promotion
APD	Root apoplastic path damage
RLD	Root length density
CONTACT	Root-soil contact loosening
RWU	Root water uptake
SEN_05	The simulated scenarios when plants with low stomatal sensitivity (simulated with halving calibrated stomatal parameters)
SEN_10	The simulation scenarios when plants with benchmarking stomatal sensitivity (simulated with calibrated stomatal parameters)
SEN_30	The simulation scenarios when plants with high stomatal sensitivity (simulated by tripling calibrated stomatal parameters)
SEN_CABA	The simulation scenarios when plants with different stomatal sensitivity to xylem ABA concentration
SEN_LEPO	The simulation scenarios when plants with different stomatal sensitivity to leaf water potential
SR-HRV	A soil-plant hydraulic model developed in this study
SWC	Soil volumetric water content
SWC50	Soil volumetric water content at a 50 cm soil depth

plant physiology, such as high-resolution X-ray computer tomography, have allowed observing the appearance of root-soil contact loosening (CONTACT, air gap formation) in drought conditions (Carminati et al., 2013; Koebernick et al., 2018). Then, when water moves through the soil-root interface, the varying permeability of root tissues and anatomical structures offer three water paths toward xylem vessels (Couvreur et al., 2018; Zarebanadkouki et al., 2019), i.e., the symplastic path via plasmodesmata across neighboring cells, the transcellular path cross membranes of neighboring cells, and the apoplastic path through intercell space and root wall. In the first two cell-to-cell paths, aquaporins, such as plasma membrane intrinsic proteins (PIPs) and tonoplast intrinsic proteins (TIPs), play a vital gating role in water molecule transportation (Maurel et al., 2015). However, drought stress usually affects the activity of aquaporins by changing hormonal regulation and gene expression (Boursiac et al., 2005; Perrone et al., 2012). Regarding the tortuous radial flow channels in the apoplastic path, constructed by apoplastic suberization and Casparian strips, these anatomical structures are prone to be destroyed with continuous root dehydration (North and Nobel, 1992; Barrios-Masias et al., 2015).

The abovementioned structural and physiological changes along the soil-root radial water path affect the hydraulic resistance of the root zone. Many studies have explored the qualitative impacts of soil-root hydraulic resistance variation on soil-plant water transport. Sade et al. (2009) and Vandeleur et al. (2009) reported the enhancement of root aquaporin activity of *Solanum Lycopersicum* and *Vitis vinifera* in drought conditions, which allows plants to maintain high transpiration and prevents leaf water potential drop-down. Carminati et al. (2013) and Koebernick et al. (2018) observed air gap formation at the soil-root interface following transpiration decrease in *Lupinus albus* L. and *Vicia faba* L.; Rodríguez-Domínguez and Brodribb (2020) detected an alleviation of hydraulic gradient enlargement in the rhizosphere of olives (*Olea europaea* L.) as a result of immediate stomatal closure. Although

these phenomena confirmed the influences of soil-root hydraulic resistance variations on the soil-plant hydraulic system, only a few simulation studies, like Cai et al. (2018) and Nguyen et al. (2020), considered the variations of root (or plant) hydraulic resistances when plants in a drought situation. The hydraulic resistance of the root system is generally assumed to be independent of the soil-root water status in most modeling applications. The conditions for the applicability of such an assumption and the related potential uncertainties in soil-plant water transport modeling are still poorly determined. Additionally, in a drought situation, the influencing factors of soil-root hydraulic resistance, such as root aquaporin activity and soil-root contact, respond simultaneously rather than independently. Other than the study by Couvreur et al. (2014b), few quantitative studies have described the effects of interactions between different soil-root hydraulic resistance variations on soil-plant water transport.

More than relying on hydraulic traits of belowground (including water potential, hydraulic resistance, and water flux in the soil-root system), water transport in the soil-plant continuum is regulated by the stomatal behavior of plants (Tardieu et al., 2017; Carminati and Javaux, 2020; Nguyen et al., 2020). According to Dixon and Joly (1895), the rhizosphere hydraulic status interacts with the stomatal behavior via the cohesion-tension mechanism. To avoid vascular embolism and desiccation, plants regulate their stomatal aperture to maintain the water balance between transpiration demand and the water supply capacity of the hydraulic system (Martin-StPaul et al., 2017). Intuitively, a lower leaf water potential stimulates stomatal closure through changes in the turgor of the guard cell. Several sources of evidence suggest that chemical compounds, such as abscisic acid (ABA) also participate in stomatal regulation (as shown in Fig. 1b) (Dodd et al., 2010; Visentin et al., 2016). According to the common definition of Tardieu et al. (1996), Tardieu et al. (1998), two types of stomatal regulation strategies can be recognized: isohydric and anisohydric. Compared with those of isohydric plants, the stomata of anisohydric plants are insensitive to hydraulic signals; therefore, the corresponding leaf water potential is usually prone to decrease with soil water depletion. However, recently, this dichotomy was challenged because it failed to describe different iso/anisohydric behavior for the same species (Schultz, 2003; Feng et al., 2019). Hochberg et al. (2018) and Ratzmann et al. (2019) argued that the classification of iso/anisohydric not only depends on the hydraulic properties of plants but also correlates with environmental conditions. Noteworthy, Carminati and Javaux (2020), using a hydraulic model, demonstrated the dominant influences of soil hydraulic conductivity loss on stomatal closure. This result implies that the soil-root zone is a key region to alter the stomatal behavior in drought conditions. However, few simulation studies have focused on the interactions of soil-root hydraulic resistance variation and stomatal regulation, especially for plants with different stomatal sensitivities.

Root water uptake (RWU) modeling is a promising tool for simulating hydraulic interactions between the rhizosphere and aboveground parts of plants (Javaux et al., 2013; dos Santos et al., 2017). In physical-based RWU models, the principle of microscopic water flow from soil to root can be analogously described by Ohm law. Gardner (1960) first described RWU as a function of the water potential gradient and hydraulic resistances between the rhizosphere and root surface. Since then, several microscopic physical RWU models have been proposed and are intended to describe the complex geometrical and morphological characteristics of the soil-root system (Cowan, 1965; Clausnitzer and Hopmans, 1994; Doussan et al., 1998; Javaux et al., 2008; Schröder et al., 2009). However, microscopic models are cumbersome to use at the field or larger scales due to the high data demands and computational costs. Thus, upscaling of RWU models to the macroscopic scale is urgently needed from a practical perspective (Couvreur et al., 2012; dos Santos et al., 2017). de Jong van Lier et al. (2008), de Jong van Lier et al. (2013) avoided developing a model based on the complex hydraulic architecture of the root system, they scaled up the single root model using mass balance theory and introduced a concept of matric flux potential to

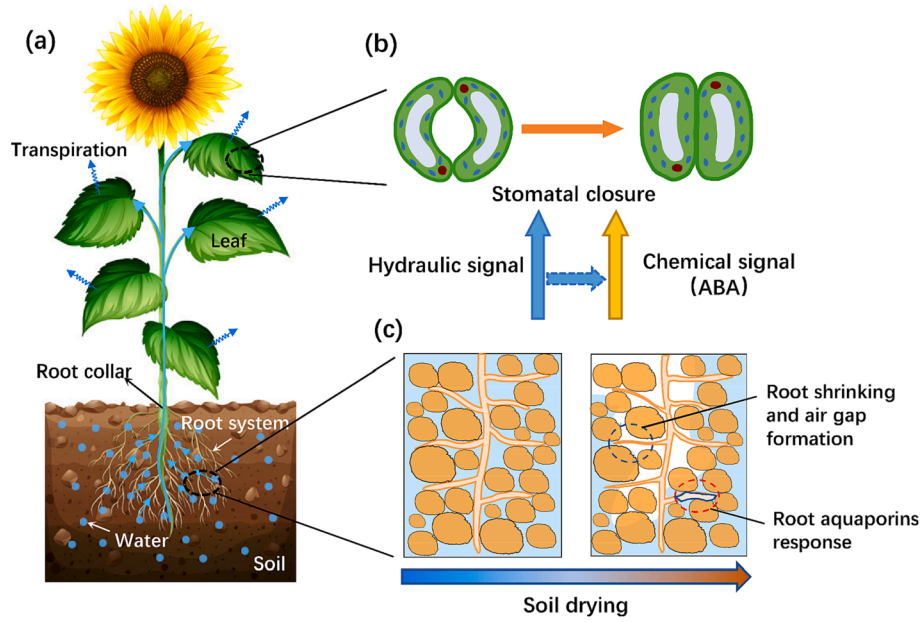


Fig. 1. Descriptions of (a) the water transport processes in a soil-plant system, (b) the stoma closing due to the stimulation of chemical (abscisic acid, ABA) and hydraulic signals produced from the root zone, (c) the root aquaporins respond in relatively wet soil, but the roots gradually shrink, and an air gap is formed at the soil-root interface in drying soil.

represent soil-plant hydraulic properties. The water flux from the soil to the root surface, the root xylem, and the leaf are systematically simulated in this model. However, for the purpose of simplification, macroscopic RWU models, including the models of [de Jong van Lier et al. \(2008\)](#), [Jong van Lier et al. \(2013\)](#) and [Couvreur et al. \(2012\)](#), [Couvreur et al. \(2014a\)](#), typically assume a constant critical potential threshold at the leaf or root collar to determine the root water absorption capacity when plants are suffering drought stress. This assumption is typically reasonable for plants with more isohydric behavior. But for anisohydric plants, their leaf water potential usually presents strong plasticity to drought stress, the usage of a constant threshold might not reflect the stomatal response to drought signal stimulations ([Huber et al., 2015](#); [Tardieu et al., 2015](#)).

To address these issues, a macroscopic water transport model SR-HRV was proposed in this study. In the SR-HRV model, the soil-root hydraulic resistance variation effects of root aquaporins promotion (AQP), apoplastic path damage (APD), and root-soil contact loosening (CONTACT) were respectively characterized by piecewise linear functions, and the collaborative regulations of hydraulic and chemical (ABA) signals on stomatal conductance (g_s) were quantified. Moreover, the parameter sensitivity analysis, calibration, and validation of the SR-HRV model were carried out based on a field experiment with sunflower plants (*Helianthus annuus* L.). The numerical experiments of plant dehydration were designed with different soil-root hydraulic resistance variations and stomatal sensitivities. The objectives of this study are to (1) determine and calibrate the sensitive parameters of the SR-HRV model, (2) clarify the individual and interactive effects of three soil-root hydraulic resistance variations (AQP, APD, and CONTACT) on water transport, and (3) identify the effects of soil-root hydraulic resistance variations on transpiration and drought signal responses for plants with different stomatal sensitivities to hydraulic and chemical signals.

2. Theory

2.1. Modeling water transport in the soil-plant system

2.1.1. Soil water movement equation

The water movement in unsaturated soil is described by the Richards

equation:

$$\frac{\partial \theta(h_s)}{\partial t} = \frac{\partial}{\partial z} \left[K_{soil}(h_s) \frac{\partial h_s}{\partial z} + K_{soil}(h_s) \right] - S(z) \quad (1)$$

where θ [$L^3 L^{-3}$], h_s [L], and K_{soil} [$L T^{-1}$] are the soil volumetric water content, water potential, and hydraulic conductivity, respectively. t [T] is the time, z [L] is the vertical coordinate (positive upward; negative downward). $S(z)$ [$L^3 L^{-3} T^{-1}$] is a sink term that accounts for RWU in this study. The transpiration (T_a [$L T^{-1}$]) is the summation of $S(z)$ in each unit depth of soil (Δz [L]) over the whole root depth (l [L]):

$$T_a = \sum_{z=0}^{z=l} S(z) \cdot \Delta z \quad (2)$$

2.1.2. Modeling water flow from the soil to root xylem by the matric flux approach

To avoid numerical instability in the RWU simulation (caused by the steep water gradient and flow line convergence near the roots), [de Jong van Lier et al. \(2006\)](#) introduced a convenient expression of matric flux potential M [L^2/T], which is defined as an integral of hydraulic conductivity $K(h)$ [$L T^{-1}$] from the permanent wilting point h_w [L] to the current potential h [L] in soil or plant tissue:

$$M = \int_{h_w}^h K(h) dh \quad (3)$$

The root system is simplified as vertically cylindrical roots with the same diameter according to [de Jong van Lier et al. \(2008\)](#), [de Jong van Lier et al. \(2013\)](#) (Fig. 2a). For a single root, as shown in Fig. 2b, the function of axisymmetric radial flow from the soil to the root surface and root xylem is written as:

$$\frac{\partial \theta}{\partial t} = -\frac{q}{r} - \frac{\partial q}{\partial r} \quad (4)$$

where q [$L T^{-1}$] is the radial flux density, r [L] is the radial distance from the root center. Based on Darcy's equation and assuming the gravitational gradients are irrelevant to the problem, q is described as:

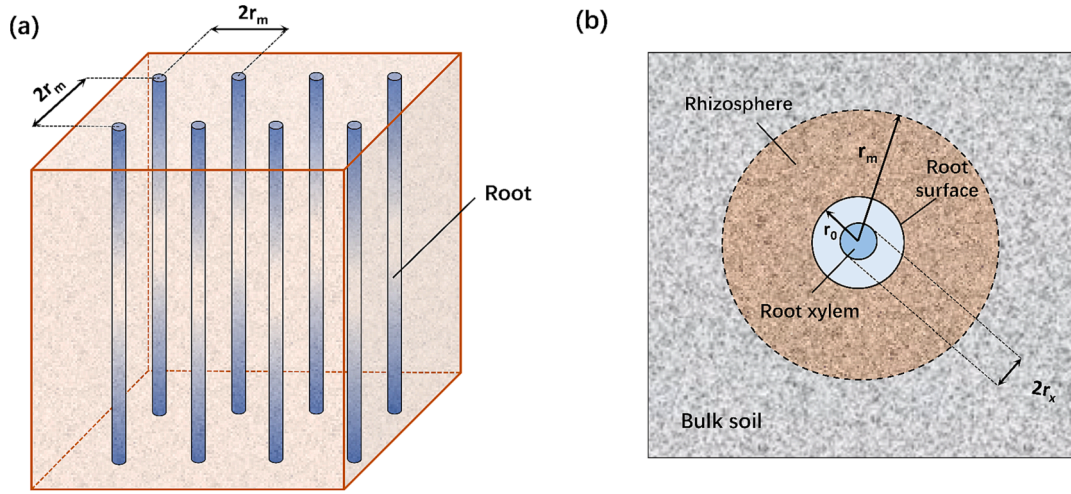


Fig. 2. Descriptions of (a) the root distribution and (b) the soil-root radial anatomical structure in the SR-HRV model (similar to the RWU model from de Jong van Lier et al. (2008), de Jong van Lier et al. (2013)).

$$q = -K \frac{\partial h}{\partial r} = -\frac{\partial M}{\partial r} \quad (5)$$

Combining Eq. (4) and Eq. (5), the radial water flow is translated into the following second-order differential equation:

$$\frac{\partial \theta}{\partial t} = \frac{\partial M}{r \partial r} + \frac{\partial^2 M}{\partial r^2} \quad (6)$$

The solution of Eq. (6) can be divided into two processes: water flows from the rhizosphere to the root surface and water flows from the root surface to the root xylem. For the soil surrounding the root, de Jong van Lier et al. (2008) developed an expression for root water uptake $S(z)$ per unit thickness of the root zone at depth z :

$$S(z) = \rho(z)[M_s(z) - M_{0s}(z)] \quad (7)$$

where $M_{0s}(z)$ [$L^2 T^{-1}$] and $M_0(z)$ [$L^2 T^{-1}$] represent the matric flux potential at the two boundaries of the rhizosphere (one boundary is bulk soil, another is root surface, as shown in Fig. 2b) at depth z , $\rho(z)$ [L^{-2}] is an integrated variable of root properties at depth z , can be calculated by:

$$\rho(z) = \frac{4}{r_0^2 - \alpha_s^2 r_m^2(z) + 2(r_m^2(z) + r_0^2) \ln[a_s r_m(z)/r_0]} \quad (8)$$

where r_0 [L] is the radius of root surface (the distance to root center), α_s represents the relative radial distance where the water content is equal to the mean soil water content (SWC) of the rhizosphere. r_0 and α_s are assumed to be 0.5 mm and 0.53 in analogy to de Jong van Lier et al. (2008). $r_m(z)$ [L] is the radial radius of the RWU, given by:

$$r_m(z) = \sqrt{\frac{1}{\pi \cdot RLD(z)}} \quad (9)$$

where $RLD(z)$ [$cm cm^{-3}$] is the root length density at depth z .

By assuming that the water content is constant within the root phloem (between the root surface and xylem) (de Jong van Lier et al., 2013), the matric flux potential relationship between the root surface and xylem can be established by solving Eq. (6):

$$M_{0r}(z) = M_x(z) + \frac{S(z)}{2} r_m^2(z) \ln \frac{r_0}{r_x} \quad (10)$$

where $M_{0r}(z)$ is the matric flux potential of the root surface based on the root radial hydraulic conductivity K_r [$L T^{-1}$]. $M_x(z)$ [$L^2 T^{-1}$] is the matric flux potential of the root xylem at depth z . The root xylem resistance is ignored, thus M_x is related to the water potential in the root collar (h_{collar} [L]). r_x [L] is the xylem radius at depth z , which is assumed

to be 0.2 mm analogous to de Jong van Lier et al. (2013).

2.1.3. Leaf water potential and stomatal aperture control by the Tardieu–Davies model

The water potentials in the leaf (h_l [L]) and root collar (h_{collar}) are correlated by the transpiration flux T_a [$L T^{-1}$] and hydraulic conductance in the stem L_{sx} [T^{-1}] according to:

$$h_{collar} = h_l + \frac{T_a}{L_{sx}} \quad (11)$$

Considering the responses of stomatal conductance (g_s [$N L^{-2} T^{-1}$]) to drought signals, Tardieu et al. (1996) proposed a model, in which g_s is controlled by the ABA concentration in the leaf xylem (C_{ABA} [$N L^{-3}$]) and the leaf water status at a leaf/plant scale:

$$g_s = g_{s,min} + \alpha \exp[C_{ABA} \beta \exp(\delta h_l)] \quad (12)$$

where $g_{s,min}$ [$N L^{-2} T^{-1}$] is the minimum stomatal conductance resulting from the cuticular conductance and the conductance through closed stomata; α [$N L^{-2} T^{-1}$], β [$N^{-1} L^3$] and δ [L^{-1}] are the influential factors on stomatal regulation, $g_{s,min} + \alpha$ represents the maximum stomatal conductance by assuming that C_{ABA} is close to 0 and h_l is very high (which can be used to calculate the potential transpiration of plants); β and δ represent stomatal sensitivity to drought signals, the greater absolute β and δ indicate the greater stomatal sensitivity to C_{ABA} and h_l , respectively.

According to Tardieu et al. (1996), C_{ABA} has a linear response to root water potential and is diluted by water flux through the plant. However, ABA is synthesized not only in roots but also in shoots (leaves) (Kurumori et al., 2018). Recent studies, according to McAdam et al. (2016), McAdam et al. (2018), Zhang et al. (2018), and Kuromori et al. (2021), etc., indicate that phloem companion cells, guard cells, especially mesophyll cells of leaves might be the potential main site of ABA synthesizing in the plant. Moreover, several root-to-shoot signals, such as CLE25 peptide and redox/calcium, have been found that regulate the ABA synthesis and stomatal responses in leaves (Takahashi et al., 2018; Yoshida and Fernie, 2018). In this study, considering the synthetic effects of the shoot and root hydraulic states on ABA synthesis, we simulated the C_{ABA} using the average water potential in the plant (h_{plant} [L]):

$$C_{ABA}(z) = a[h_{plant} + c]/[T_a/m + b] \quad (13)$$

where a [$N T^{-1} L^{-1}$], b [$L^3 T^{-1}$], and c [L] are fitting parameters; and m [plants L^{-2}] is the plant density used for transferring T_a from the field scale ($L^3 L^{-3} T^{-1}$) to the plant scale ($L^2 T^{-1}$) (in this study, m equal to 4

plants m^{-2}). Regarding h_{plant} , we assumed it is equal to the average water potentials in the leaf and root, the root water potential indicates the average water potentials in the root surface and xylem across the whole root system.

At the field scale, plant transpiration (T_a) can be calculated based on the stomatal conductance at the canopy scale, which is estimated by upscaling g_s through weighting by the effective leaf area index (LAI_{eff}) based on big-leaf theory (Ding et al., 2014):

$$T_a = \frac{s\phi_n + \rho_a c_p g_a D}{\lambda(s + \gamma g_a / (g_s \text{LAI}_{\text{eff}}))} \quad (14)$$

where s [$\text{ML}^{-1}\text{T}^{-2}\text{K}^{-1}$] is the slope of the curve representing the saturation vapor pressure vs temperature relationship; ρ_a [ML^{-3}] is air density; c_p is the specific heat of air; ϕ_n [MT^{-3}] is net radiation; G_a [LT^{-1}] is aerodynamic conductance at the plant level; D [LMT^{-2}] is the air vapor pressure deficit; λ [L^2T^{-2}] is the latent heat of vaporization; γ [$\text{L}^{-1}\text{MT}^{-2}\text{K}^{-1}$] is the psychrometric constant. Where LAI_{eff} can be estimated from the actual leaf area index (LAI) according to Bastiaanssen et al. (2012):

$$\text{LAI}_{\text{eff}} = \frac{\text{LAI}}{0.3\text{LAI} + 1.2} \quad (15)$$

With meteorological data, leaf area index, plant height, and G_s , the plant transpiration can be calculated at the field scale based on the Penman-Monteith direct equation in the SWAP (Soil-Water-Atmosphere-Plant) model (Kroes et al., 2017).

2.2. Hydraulic resistance responses of the soil-root system in water-limited conditions

2.2.1. Hydraulic conductivity reduction in the rhizosphere due to CONTACT

The hydraulic conductivity of the rhizosphere is critical for plant water use efficiency. However, in drought conditions, the root surface detaches from the soil accompanied by the root cells begin to shrink and reduce the secretion of hydrophilic substances. Carminati et al. (2013) observed an air gap that appears along with the taproot when h_s ranged from -0.005 to -0.01 MPa and extended to 0.35 mm when h_s decreased below -0.04 MPa. Koebernick et al. (2018) found that the root-soil contact of the taproot and lateral roots of *Vicia faba* L. significantly decreased when the h_s decreased below -0.02 MPa. Once an air gap is formed, the hydraulic conductivity of the rhizosphere is extremely limited because water transport mainly relies on gas diffusion. Liu et al. (2015) detected an increasing air gap width with h_s decreasing, which increased the hydraulic resistance at the root-soil interface for *Robinia pseudoacacia*. Few simulations had quantified the effects of air gap formation on root-soil interface hydraulic resistance and root water uptake (Liu et al., 2015; de Willigen et al., 2018), however, complex model structures and numerous parameters were introduced in these simulations. In this study, to limit model complexity, based on the changing trend of the quantitative relationship between water potential and hydraulic conductivity in the rhizosphere, similar to Couvreur et al. (2014b), we used a piecewise linear function to describe the effects of CONTACT:

$$K_{s-r} = \alpha_{\text{CONTACT}}^{s-r} K_{\text{soil}} \quad (16)$$

$$\alpha_{\text{CONTACT}}^{s-r} = \begin{cases} 1 & h_s - r \geq h_{sr,1} \\ 1 - \frac{h_s - r - h_{sr,1}}{h_{sr,2} - h_{sr,1}} & h_{sr,2} \leq h_s - r < h_{sr,1} \\ 0 & h_s - r \leq h_{sr,2} \end{cases} \quad (17)$$

where the hydraulic conductivity in the rhizosphere (K_{s-r} [L T^{-1}]) is modified as a function of $\alpha_{\text{CONTACT}}^{s-r}$, $\alpha_{\text{CONTACT}}^{s-r}$ is correlated with the water

potential in the rhizosphere (h_{s-r} [L]). Here, $h_{sr,1}$ [L] and $h_{sr,2}$ [L] are the two thresholds of water potential in the rhizosphere. K_{s-r} is equal to the hydraulic conductivity in bulk soil (K_{soil}) without considering the CONTACT effects. However, once CONTACT occurs, K_{s-r} linearly decreases when h_{s-r} is lower than $h_{sr,1}$, and K_{s-r} decreases to 0 when h_{s-r} drops to $h_{sr,2}$. The matric flux potentials in the boundaries of the rhizosphere (Eq. (7)) were modified by K_{s-r} in the bulk soil and root surface.

2.2.2. Variations in root radial conductivity due to AQP and APD

The root radial conductivity (K_{rr} [L T^{-1}]) regulates the water flow from the root surface to the xylem, which has been experimentally found to be much lower than the soil conductivity in wet soil (soil water potential above -0.1 – -0.01 MPa) (Draye et al., 2010). The permeability of plasma membranes determines the hydraulic resistance in the trans-cellular and symplastic radial pathways, while it is reversibly regulated by the activity of aquaporins. Many pieces of evidence show that aquaporins play significant roles in the response to environmental stimuli (Pawlowicz and Masajada, 2019), but inconsistent impacts of aquaporins response on root radial conductivity have been found in a variety of plants (Boursiac et al., 2005; Zhao et al., 2008; Perrone et al., 2012). Might be mediated by ABA, plants can enhance their root extraction capacity when suffering slight drought stress (Quintero et al., 1999; Hose et al., 2000; Hachez et al., 2012; Pawlowicz and Masajada, 2019). Glinka (1980) and Quintero et al. (1999) found that the root hydraulic conductivity of sunflower increased by 150–250 % when supplied with a $4 \mu\text{M}$ external ABA solution. In this study, similar to Couvreur et al. (2014b), we adopted a multiplicative factor (α_{AQP}) to quantify the promotion effects of AQP on K_{rr} , which is linearly correlated with the water potential in root tissue (h_{root}), expressed as:

$$\alpha_{\text{AQP}} = \begin{cases} 1 & h_{\text{root}} \geq h_{\text{root},1} \\ 1 + \frac{h_{\text{root}} - h_{\text{root},1}}{h_{\text{root},3} - h_{\text{root},1}} \times R & h_{\text{root},3} \leq h_{\text{root}} < h_{\text{root},1} \\ 1 + R & h_{\text{root}} < h_{\text{root},3} \end{cases} \quad (18)$$

where h_{root} [L] is assumed to be the average water potential of the root surface and xylem at a specific depth; $h_{\text{root},1}$ [L] and $h_{\text{root},3}$ [L] are two thresholds of water potential in the root tissue, R is the maximum promotion factor of K_{rr} due to the AQP effects, K_{rr} linearly increases starting at $h_{\text{root}} < h_{\text{root},1}$, and increases up to $(1 + R)$ times K_{rr}^* when $h_{\text{root}} < h_{\text{root},3}$. Different from Couvreur et al. (2014b), we selected h_{root} rather than h_s as the influential factor to quantify the promotion of K_{rr} , which could better reflect the hydraulic status in roots, especially when a large water potential gradient exists between soil and roots.

In addition to the cell-to-cell pathway, apoplastic transport is another important radial water pathway in root tissue. The smoothness of apoplastic water flow is dominantly associated with the root anatomical structure (Barrios-Masias et al., 2015). However, continuous dehydration usually damages the root anatomical structure. Cuneo et al. (2016) observed that the mechanical damage of cortical cells (lacunae formation) coincided with root shrinkage of grapevines (*Vitis berlandieri* X.) when the stem potential was less than -0.6 MPa. Tataranni et al. (2015) found that the number of suberized cellular layers of olive (*O. europaea* L.) increased to nearly-five from the endodermis to the cortex, and the reduction in radial hydraulic conductivity exceeded 80 % when the leaf water potential was lower than -6.5 MPa. However, although several correlations regarding microcosmic morpho-anatomical changes and root hydraulic conductivity were proposed for specific plants (Tataranni et al., 2015), no robust models have been established that can effectively describe the APD effects on the root hydraulic status. For the purpose of simplification, according to the potential influences of APD on K_{rr} , we introduced a linear function to characterize the effects of APD based on h_{root} , as expressed in Eq. (19).

$$\alpha_{APD} = \begin{cases} 1 & h_{root} \geq h_{root,2} \\ 1 - \frac{h_{root} - h_{root,2}}{h_{root,4} - h_{root,2}} & h_{root,2} < h_{root} < h_{root,4} \\ 0 & h_{root} < h_{root,2} \end{cases} \quad (19)$$

where $h_{root,2}$ [L] and $h_{root,4}$ [L] are the two thresholds of water potential in the root tissue, K_{rr} linearly decreases when h_{root} is below $h_{root,2}$, and we considered that radial water flow stops if h_{root} decreases to $h_{root,4}$.

Then, a multiplicative relationship is used to describe the combined effects of α_{AQP} and α_{APD} as expressed in Eq. (20):

$$K_{rr} = \alpha_{AQP} \cdot \alpha_{APD} \cdot K_{rr}^* \quad (20)$$

3. Methodology

3.1. Model coupling and implementation

In the SR-HRV model, the water transport processes in the soil-vegetation-atmosphere continuum are subdivided into water movement from the rhizosphere to the root surface, root xylem, and leaf stomata (described in the **Theory** section). We originally compiled these processes based on the “rootextraction” subroutine of the SWAP model (Kroes et al., 2017), which can provide the numerical solution of the Richards equation (Eq. (1)). Based on the physical RWU model developed by de Jong van Lier et al. (2013), we embedded the Tardieu–Davies model and three piecewise linear functions in the calculation procedure to describe the effects of stomatal regulation and soil-root hydraulic resistance variations (described in Section 2.2). The water potentials in the leaves and roots were implicitly correlated with the water potential in the root collar (h_{collar}). The Newton-Raphson iteration of h_{collar} was carried out until the simulated transpiration of the RWU model and stomatal conductance model is converged.

3.2. Parameter sensitivity analysis, calibration, and validation

Based on field observation of sunflower plants, we analyzed the parameter sensitivities of the SR-HRV model and selected the sensitive parameters for calibration and validation. The initial soil water distribution, daily meteorological data, groundwater table, and crop morphological indicators (including crop height, LAI, and RLD) were collected as the input variables to run the SR-HRV model. A soil profile (0–300 cm) was discretized into 2 cm layers, the upper and bottom boundaries of the soil profile were controlled by atmospheric conditions and groundwater tables, respectively. The details of the field experiment measurements, parameter sensitivity analysis, and calibration methods are described in this section.

3.2.1. Field experiments

The field experiments comprised three plots of Y2016_1, Y2016_2, and Y2017, each plot with an area of approximately 30 m² at the Yichang experimental station in the Hetao Irrigation District in Inner Mongolia, China (108°0′15.3″E, 41°4′2.16″N). Sunflowers (*Helianthus annuus* L., cultivar JK/GL601) were grown from 5 June to 19 September 2016 and from 27 May to 15 September 2017. The row distance between the sowing points was 50 cm. The field was irrigated (250–300 mm) before sowing (on 9 May 2016 and 15 May 2017), and no irrigation was applied during the plant growth period.

The meteorological data of the experimental site were retrieved from an automatic meteorological station located 10 m from the experimental site, the groundwater table was monitored every 1–5 days in a water well near the field. SWCs (0–100 cm) were measured by the oven drying and weighing method at 1, 27, 51, 69, and 105 days after sowing in 2016 and at 1, 30, 60, 70, 80, and 112 days after sowing in 2017. The soil samples were collected every 10 cm down to 40 cm depth and from there every 20 cm down to 100 cm depth. The van Genuchten (1980) of the

soils (Table 1) were initially estimated by the Rosetta pedotransfer functions (Schaap et al., 2001) and calibrated (using PEST (Parameter Estimation software)) with the first two soil water measurements (0–100 cm soil profile) before 30 days after sowing of the two years (no crop was assumed due to the low transpiration during this period). The stomatal conductance (g_s) of the youngest fully expanded leaf of tagged plants was measured using a portable photosynthetic apparatus (LI-6400, Li-Cor, Lincoln, USA) at midday (from 10 a.m. to 12 a.m.) on windless and sunny days (42, 53, 65, 72, and 107 days after sowing in 2016; 38, 43, and 69 days after sowing in 2017). Each leaf was measured 5 times. The length (L) and width (W) of green leaves and the height of three representative plants were measured with a tape starting from 27 and 22 days after sowing in 2016 and 2017, respectively (Fig. 3a and b). The leaf area (LA) was calculated based on the empirical relationship ($LA = 0.6564 \times L \times W$) proposed by Chen (1984). The LAI was defined as the green leaf area per unit of ground surface area.

The dynamic root distributions of the plants were monitored by a minirhizotron and ROOT 700 system (Ecomatik, Munich, Germany). Before sowing, four parallel transparent rhizotubes, with an interval of 1 m horizontally, were diagonally inserted into the soil in the middle of the two sowing rows, the insertion angle of each rhizotubes was 45° to the horizontal, and the vertical depth of insertion was 1 m. The roots surrounding the rhizotubes were scanned with a ROOT 700 digital camera layer by layer, the vertical height of each scan layer was controlled to 15 cm. The root length of the sunflower in each scanned image was distinguished and calculated with the root analysis software WinRHIZOTRON (Regent Instruments Inc., Quebec, Canada), the RLD was converted by assuming that the equivalent soil thickness surrounding the rhizotubes was equal to 2.5 mm. The RLD of the whole field was determined by the average RLD status in four parallel rhizotube images. The RLD of three plots at different soil depths and sampling dates are shown in Fig. 3c.

3.2.2. Base values of the hydraulic properties in the soil-plant system

The base values of the hydrophysiological parameters in the SR-HRV model are listed in Table 2. The radial hydraulic conductivity K_{rr}^* [L T⁻¹] was transformed by the radial root hydraulic conductance L_{rr} [T⁻¹] according to the conversion equation introduced by de Jong van Lier et al. (2013). The L_{rr} of sunflower was investigated by Adiredjo et al. (2014) and Liu and Shan (2003). We took the average upper limit of their measurements 2.08×10^{-6} d⁻¹ and 2.97×10^{-6} d⁻¹ as the base value of L_{rr} . By assuming K_{sx} was constant in the xylem of stems and leaves at a 1 m distance from the root collar to the evaporation site of the leaves, the stem-leaf hydraulic conductance L_{sx} [T⁻¹] of plants was transformed by stem-leaf hydraulic conductivity K_{sx} [L T⁻¹] according to Ahmad et al. (2018) and Gleason et al. (2017).

The base values of the stomatal parameters regarding C_{ABA} and g_s (Eqs. (12) and (13)) were referred to the experimental results of Tardieu et al. (1996). δ controls the participation activity of h_i in stomatal regulation; as absolute δ increases, the regulation capacity of h_i to g_s is enhanced. According to Tardieu et al. (1996), Tardieu et al. (2015), the reference δ values are 0 for sunflower; however, considering drought tolerance of the sunflower and to prevent serious xylem cavitation (Stiller and Sperry, 2002; Ahmad et al., 2018; Cardoso et al., 2018), we set the base δ as −0.2.

Referring to the simulation of Couvreur et al. (2014b), we assumed that soil-root contact loosening occurs when h_{s-r} equals −0.05 MPa and K_{s-r} decreases to 0 when h_{s-r} decreases to −1.5 MPa due to incomplete soil-root contact (a hypothetical state, the water absorption is ignored if h_{s-r} below $h_{sr,2}$); regarding the function of aquaporin responses, based on the related experimental and simulation studies in Section 2.2.2, we assumed that the K_{rr} would be subject to an increase of 200 % ($R = 2$) when the h_{root} changes from −0.005 MPa to −0.1 MPa. Considering the correlation between APD and CONTACT, we preliminarily set the same base water potential thresholds for APD and CONTACT.

Table 1
Soil texture and van Genuchten parameters in field experiments.

Soil depth cm	Soil texture			van Genuchten parameters				
	Clay %	Loam %	Sand %	θ_r $\text{cm}^3 \text{cm}^{-3}$	θ_s $\text{cm}^3 \text{cm}^{-3}$	K_s cm d^{-1}	α cm^{-1}	n -
0–10	11.65	81.07	7.28	0.0663	0.4198	14.808	0.0259	1.3602
10–100	9.09	75.74	15.17	0.0733	0.4609	15.425	0.0194	1.3602
100–300	–	–	–	0.0698	0.5830	0.648	0.0096	1.6949

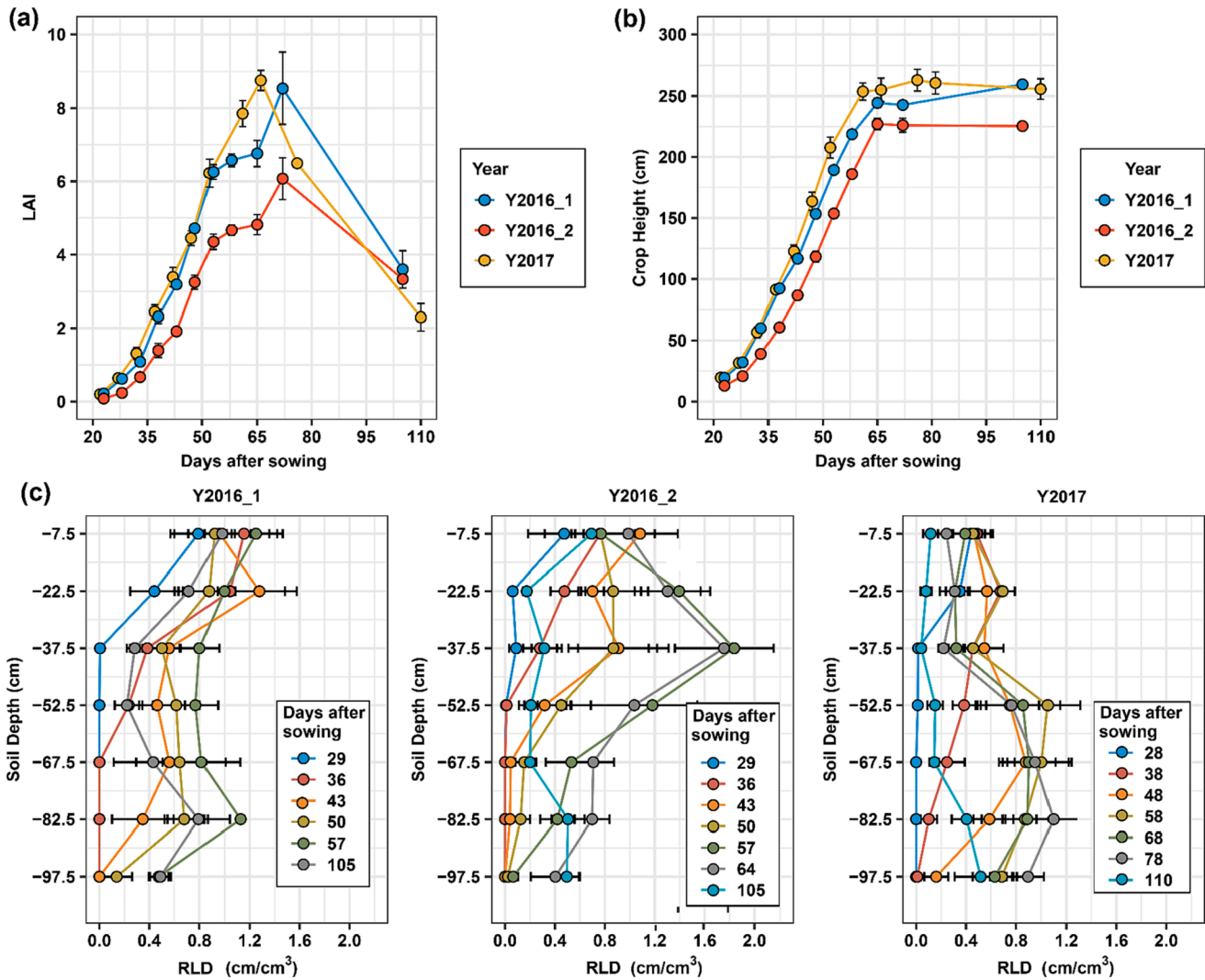


Fig. 3. Dynamic leaf area index (LAI, m^2/m^2) (a), crop height (cm) (b), and root length density (RLD, cm cm^{-3}) (c) of sunflower in the Y2016_1, Y2016_2, and Y2017 field experiments.

3.2.3. Sensitivity analysis

The effects of soil-root hydraulic resistance variations and stomatal regulation are quantified by different hydrophysiological parameters in the SR-HRV model. To test the sensitivity of simulated g_s and SWC to different parameters (based on field observations of Y2016_1), a classic sensitivity analysis method of Morris (1991) was used in this study. This method, developed from one step at a time (OAT) theory, can effectively balance the efficiency and accuracy of the calculation. Regarding a model with k input parameters, two sets of input parameters X and X^* are determined:

$$X = x_1, x_1 \dots x_{i-1}, x_i, x_{i+1}, \dots, x_k, X^* = x_1, x_1 \dots x_{i-1}, x_i + \Delta, x_{i+1}, \dots, x_k \quad (21)$$

where the same values in X and X^* are selected except for the i_{th} parameter, a variation step Δ is added to the i_{th} parameter in X^* , Δ is between $1/(p-1)$ and $1-1/(p-1)$ times the variation range for the i_{th} parameter, and p represents the change gradients of the i_{th} parameter. The elementary effects $f_i(X)$ of the i_{th} parameter can be characterized as the change rate of model output $Y(X)$ and $Y(X^*)$:

$$f_i(X) = \frac{Y(X^*) - Y(X)}{\Delta} \quad (22)$$

To avoid the randomness of parameter values, the element effects for each input parameter are repeatedly estimated r times by randomly sampling $X^{(1)}, X^{(2)}, \dots, X^{(r)}$. Two evaluation indexes μ_i and σ_i are defined as the mean and standard deviation of the distribution of the elementary

Table 2

Base values of hydraulic parameters, stomatal parameters, and soil-root hydraulic resistance response parameters in the SR-HRV model for sunflower.

Types	Parameter	Unit	Base value	References
Hydraulic conductivity/conductance	K_{tr}^*	cm d^{-1}	2.5235×10^{-6}	Transformed by L_{tr} in Adiredjo et al. (2014) and Liu and Shan (2003)
	L_{sx}	d^{-1}	1.0800×10^{-4}	Transformed by K_{sx} in Ahmad et al. (2018) and Gleason et al. (2017)
Stomatal parameters	$g_{s,min}$	m/s	0.0002	Modified based on Tardieu et al. (1996)
	α	$\text{mol m}^{-2} \text{s}^{-1}$	2.1980	Tardieu et al. (1996)
	β	$\mu\text{mol}^{-1} \text{m}^3$	-0.0063	Tardieu et al. (1996)
	δ	MPa^{-1}	-0.2000	Tardieu et al. (1996)
	a	$\text{pmol s}^{-1} \text{MPa}^{-1} \text{plant}^{-1}$	-8.2750	Tardieu et al. (1996)
	b	$\text{mm}^3 \text{s}^{-1} \text{plant}^{-1}$	35.000	Tardieu et al. (1996)
	c	MPa	-0.1800	Tardieu et al. (1996)
Soil-root hydraulic resistance response parameters	$h_{sr,1}$	MPa	-0.0500	Couvreux et al. (2014b)
	$h_{sr,2}$	MPa	-1.5000	Couvreux et al. (2014b)
	$h_{root,2}$	MPa	-0.0500	Assumed
	$h_{root,4}$	MPa	-1.5000	Assumed
	$h_{root,1}$	MPa	-0.0050	Couvreux et al. (2014b)
	$h_{root,3}$	MPa	-0.100	Couvreux et al. (2014b)
	R	-	2.0000	Couvreux et al. (2014b)

effects of the i_{th} parameter:

$$\mu_i = \frac{1}{r} \sum_{j=1}^r |f_i(X^{(j)})|, \sigma_i = \sqrt{\frac{1}{(r-1)} \sum_{j=1}^r (f_i(X^{(j)}) - \mu_i)^2} \quad (23)$$

where μ_i can assess the overall influence of an input i_{th} parameter on the model output, and σ_i can describe nonlinear effects and interactions between the i_{th} and other parameters.

In this study, to alleviate the influences of meteorology, groundwater table, and crop growth, the cumulative values of μ_i and σ_i ($C-\mu$ and $C-\sigma$) were selected to evaluate the dynamic sensitivity responses of g_s and SWC to seven stomatal regulation parameters and seven soil-root hydraulic resistance variation parameters (Table 2). To avoid the disturbance of soil evaporation, rainfall, and groundwater table changes on soil water dynamics, the SWC at a 50 cm soil depth (SWC50) was selected in the sensitivity analysis. Nine levels between the ranges of 0.5–1.5 times the base values and 10 times ($r = 10$) random samplings were set for each parameter. The input parameter dataset generation and evaluation index calculation were based on Simlab software (Tarantola and Becker, 2017).

3.2.4. Calibration and validation of sensitive parameters

To reduce the parameter uncertainty, the plant hydraulic conductivity/conductance, sensitive stomatal parameters, and sensitive soil-root hydraulic resistance variation parameters were selected to cali-

brate (Y2016_1 and Y2017) and validate (Y2016_2) based on field experiments with sunflower plants. In parameter calibration, PEST software (White et al., 2020) was used in this study, which combines the fast convergence speed of the Gauss-Newton method with the global search characteristics of the gradient descent method, which can improve parameter adjustment efficiency and avoid the subjectivity of artificial parameter adjustment (Doherty, 2004). The average weighted square relative difference F_j between measured and simulated g_s (converted to the daily average from the observation at midday based on the sine function) and SWCs at seven depths were collected as the objective function, the parameters can be calibrated by multiple stepwise iterative methods:

$$\min : F_j = \frac{1}{m} \sum_{i=1}^m \left[\gamma \left(1 - Y_{ij}^s / Y_i^o \right) \right]^2 \quad (24)$$

where Y_i^o is the i_{th} observed value, Y_{ij}^s is the i_{th} simulated value in the j_{th} optimization, m is the total number of observations, and γ is the weight of the i_{th} observation. In this study, considering the differences in the number of observations of g_s (1) and SWCs (7) during a day, the weights of relative square differences of g_s were set as 7 times those of the SWCs.

3.3. Simulation scenario design

To better focus on soil-root hydraulic resistance variation effects and their interaction with the stomatal behavior, two types of ideal drying-out scenarios were designed in our study.

Simulation Scenario I: With the calibrated stomatal parameters, the individual and interactive effects of three soil-root hydraulic resistance variations (AQP, APD, and CONTACT) were investigated. Specifically, five soil-root hydraulic resistance scenarios, i.e., (i) Constant scenario (without considering the root-soil hydraulic resistance variations), the variable soil-root hydraulic resistance variation scenarios with the individual effects of the (ii) root aquaporins promotion (AQP), (iii) apoplastic path damage (APD), (iv) root-soil contact loosening (CONTACT), and (v) their interactive effects (AQP + APD + CONTACT, A + A + C) on the soil-plant water potentials and RWU distribution were simulated, respectively.

Simulation Scenario II: We expanded the calibrated stomatal regulations (benchmarking stomatal sensitivity, SEN_10), to assume that plants have higher (SEN_30, more isohydric) and lower (SEN_05, more anisohydric) stomatal sensitivities to leaf water potential (SEN_LEPO) and xylem ABA concentration (SEN_CABA), which were simulated by halving and tripling the calibrated stomatal parameters β and δ (the stomatal sensitivity parameters regarding C_{ABA} and h_l , as shown in Eq. (12)). The effects of three soil-root hydraulic resistance variations (AQP, APD, and CONTACT) on plant transpiration and drought signal responses were compared when plants with different stomatal sensitivities.

In Simulation Scenarios I and II, a soil profile (0–100 cm), vertically discretized in 1 cm layers, was established. The upper and bottom boundaries were controlled by constant atmospheric and free drainage conditions. The soil physical properties were the same as those in the field experiment (Table 1). The atmospheric conditions and plant growth status, including LAI, plant height, and RLD distribution, were based on the measurements on 8 August 2016, when the sunflower development stage was equal to 1.0. The initial soil water pressure in each compartment was in hydrostatic equilibrium with the groundwater table at a depth of 5 m. No rain, irrigation, or soil evaporation was assumed, the soil water consumption or storage in the soil profile depended only on RWU and bottom drainage.

4. Results and discussion

4.1. Sensitivity analysis of the hydrophysiological parameters

The cumulative sensitivity indexes (C_μ and C_σ) of simulated g_s and SWC50 to the hydrophysiological parameters of the SR-HRV model are shown in Fig. 4.

The C_μ and C_σ of simulated g_s response to stomatal regulation parameters (α , β , and δ) and xylem ABA concentration parameters (a , b , and c) are increased earlier and greater than to other parameters (Fig. 4a and b). Especially, the C_μ and C_σ of g_s response to ABA-related parameters (β , a , b , and c) increase from 0.01 m s^{-1} to 1.0 m s^{-1} , which are obviously higher than to soil-root hydraulic resistance variation parameters. These results might imply a response mechanism of the plants to drought stress, that is the stomata actively respond before the occurrence of soil-root hydraulic resistance variations. This phenomenon has been reported by Carminati et al. (2013) and Koebernick et al. (2018) for white lupins and *Vicia faba* L. The reason behind this is that the plants could reduce the stomatal aperture with the dual stimulations of hydraulic and chemical drought signals, which effectively prevent water loss and water potential drop-down around roots (Faiz and Weatherley, 1982; Carminati et al., 2020; Rodriguez-Dominguez and Brodribb, 2020). For some plants with more anisohydric behaviors, such

as sunflowers, g_s is insensitive to hydraulic signals (Tardieu and Simonneau, 1998; Cardoso et al., 2020), ABA-related processes would play a more essential role in regulating the stomatal behavior once plants suffer drought stress.

Moreover, compared to APD ($h_{\text{root},2}$ and $h_{\text{root},4}$) and CONTACT ($h_{\text{sr},1}$ and $h_{\text{sr},2}$) parameters, the C_μ and C_σ of simulated g_s to AQP parameters (R , $h_{\text{root},1}$, and $h_{\text{root},3}$) are responded earlier and are generally greater before the 30th day. This is attributed to the AQP effects being easily triggered in the simulation ($h_{\text{root}} < -0.1 \text{ MPa}$) even when the soil is relatively wet. In this situation, the increase in K_{tr} can effectively reduce the hydraulic resistance of the whole water transport system (Sade et al., 2009; Chaumont and Tyerman, 2014). However, after the 30th and 60th days, the C_μ and C_σ response to APD ($h_{\text{root},2}$ and $h_{\text{root},4}$) and CONTACT ($h_{\text{sr},1}$ and $h_{\text{sr},2}$) parameters rapidly increase (nearly 2–5 orders of magnitude), even approach to the C_μ and C_σ of g_s response to AQP parameters after 90th days, which means that APD and CONTACT began to dominantly regulate g_s during this period (as the explanation in Section 4.3). These results are well responded to recent outstanding works from Carminati and Javaux (2020) and Cai et al. (2021), who verified that soil-root hydraulic properties play a vital role in the soil-plant water transport system.

In addition, the response orders of the cumulative sensitivity indexes (C_μ and C_σ) of simulated SWC50 to different parameters are similar to

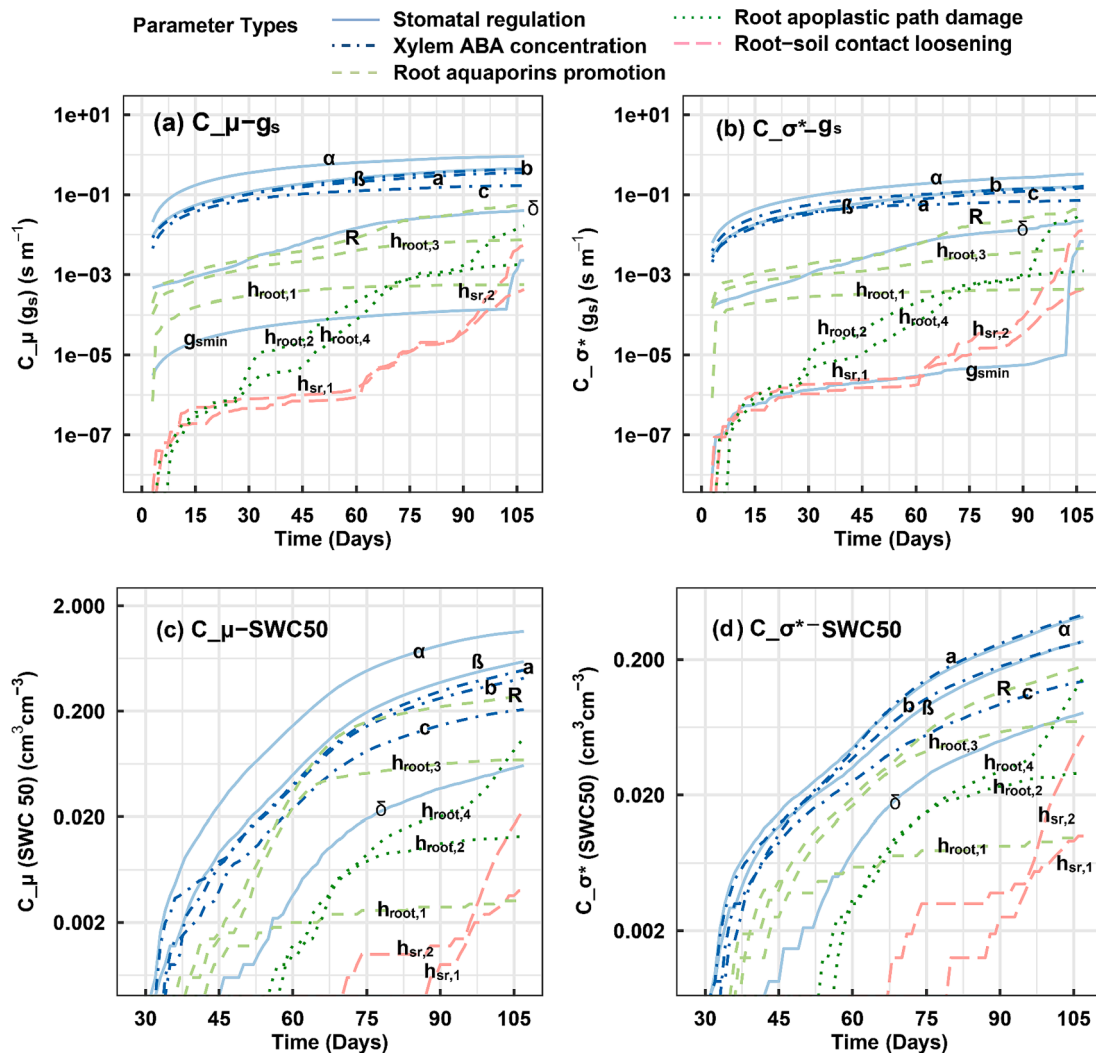


Fig. 4. Daily cumulative evaluation indexes of (a) C_μ and (b) C_σ from the Morris sensitivity analysis of stomatal conductance (g_s) and soil water content at a soil depth of 50 cm (SWC50) from the 1st to the 107th day in 2016 with respect to the parameters of stomatal regulation and hydraulic resistance variation in the soil-root system.

those of g_s (Fig. 4c and d). This implies a close link between stomatal regulation and soil water status in the SR-HRV model. A greater g_s induces higher transpiration and more negative water potential in plants, thereby accelerating water consumption from the soil. In contrast, the continuous decrease in soil moisture prompts a considerable decrease in the water potential and hydraulic conductivity of the rhizosphere, thereby stimulating drought signal production and stomatal closure. The hydrophysiological parameters collectively participate in the hydraulic interaction between the stomata and soil. Thus, the C- μ and C- σ of g_s and SWC50 to hydrophysiological parameters are synchronously responded. However, the parameter sensitivity might be different when plants with various hydrophysiological traits (e.g., plant morphology, stomatal sensitivity, drought signal generation, and resistance to soil-root hydraulic resistance variations) and growth conditions (e.g., vapor pressure deficit and soil environments). For example, the soil hydraulic properties significantly affect water uptake patterns and soil-plant hydraulics in drought situations (Cai et al., 2021), the sensitivity of plant hydraulic traits (including water potential, resistance, and water flux in the different organs or tissues of the plant) to CONTACT and APD parameters in sandy (or loam) soil is probably higher than those in clay soil because CONTACT and APD are more easily triggered. Therefore, it is necessary to test the parameter sensitivity in specific scenarios before the model application.

4.2. Calibration and validation of hydrophysiological parameters

In this section, besides L_{sx} and K_{rr}^* , the sensitive hydrophysiological parameters, including three stomatal regulation parameters (α , β , and δ), and the most sensitive parameter in each hydrophysiological process of C_{ABA} response, AQP, APD, and CONTACT (b , R , $h_{root,4}$, and $h_{sr,2}$), were selected for calibration (Y2016_1 and Y2017) and validation (Y2016_2) using field observations of g_s and SWCs. The calibrated parameters are shown in Table 3.

Fig. 5b and d show the calibrated simulation results of g_s and SWCs in Y2016_1 and Y2017. The g_s rapidly decreased after the 50th day in Y2016_1, and eventually dropped down to below 0.001 m s^{-1} after the 100th day. Differently, in Y2017, g_s decreased slightly from the 25th to the 50th day and then remained relatively stable at $0.01\text{--}0.0125 \text{ m s}^{-1}$. The SWCs (0–100 cm depth) in Y2016_1 decreased from 0.32 to $0.43 \text{ cm}^3 \text{ cm}^{-3}$ to $0.13\text{--}0.15 \text{ cm}^3 \text{ cm}^{-3}$ during the whole simulation period, while the SWCs in Y2017 maintained a higher status of $0.19\text{--}0.26 \text{ cm}^3 \text{ cm}^{-3}$ at a soil depth of 50–90 cm. These results indicate that the sunflower experienced varying degrees of drought stress in Y2016_1 and Y2017. More severe drought stress took place in Y2016_1, which mainly resulted from the rapid decline in the groundwater table in 2016 (in Fig. 5a). Similar to Y2016_1, large declines of g_s and SWCs are simulated in Y2016_2 for validation. The simulated K_{s-r} in the validation rapidly decreased after the 80th day (below K_{soil}) and eventually dropped to $4 \times 10^{-8} \text{ cm d}^{-1}$ after the 90th day (Fig. 5c), which indicates that the CONTACT effects were activated during this simulation period.

Table 3

Calibrated sensitive hydraulic parameters of the SR-HRV model for sunflower.

Parameters	Unit	Calibrated values	95 % percent confidence limits	
			lower limit	upper limit
L_{stem}	d^{-1}	1.52×10^{-4}	7.23×10^{-5}	2.32×10^{-4}
K_{rr}^*	cm d^{-1}	3.79×10^{-6}	2.45×10^{-6}	5.13×10^{-6}
α	$\text{pmol s}^{-1} \text{ MPa}^{-1}$	2.21	1.76	2.66
β	plant^{-1}			
	$\mu\text{mol}^{-1} \text{ m}^3$	-4.59×10^{-3}	-7.60×10^{-3}	-1.58×10^{-3}
δ	MPa^{-1}	-0.22	-0.39	-0.05
b	$\text{mm}^3 \text{ s}^{-1} \text{ plant}^{-1}$	39.98	18.08	78.16
R	–	2.60	1.35	3.85
$h_{sr,2}$	MPa	-1.22	-2.13	-0.31
$h_{root,4}$	MPa	-1.77	-2.75	-0.79

Moreover, influenced by the AQP effects, the K_{rr} in the validation successively increased over $9.0 \times 10^{-6} \text{ cm d}^{-1}$ from the topsoil to deeper soil from the 40th to the 100th day. However, due to the APD effects, K_{rr} gradually decreased after the 85th day.

The g_s and SWCs are two critical hydraulic indicators of the soil-plant system, which affect the water flow and drought signal generation in the leaf and rhizosphere (Carminati et al., 2020; Cai et al., 2021). In the model calibration and validation, we obtained relatively good agreements between simulations and observations for g_s (root mean square error (RMSE) of $5.56 \times 10^{-4} - 1.19 \times 10^{-3} \text{ m/s}$, determination coefficient (R^2) of 0.87–0.98) and SWCs (RMSE of $0.008\text{--}0.0511 \text{ cm}^3 \text{ cm}^{-3}$, R^2 of 0.70–0.99). Moreover, similar to the experimental results from Tardieu et al. (1996), Tardieu et al. (1998), who observed g_s decreased from 0.053 m s^{-1} to 0.002 m s^{-1} (at the hourly scale) as h_i decreased from -0.9 MPa to -2.5 MPa and C_{ABA} increased from 0 to $425 \mu\text{mol m}^{-3}$ for sunflower. We simulated g_s decreased from 0.013 m s^{-1} to 0.001 m s^{-1} at the diurnal scale (approximately from 0.041 m s^{-1} to 0.003 m s^{-1} at midday based on a sine function) company with the h_i decreased from -0.01 MPa to -1.7 MPa and C_{ABA} increased from 48 to $423 \mu\text{mol m}^{-3}$. Considering the genetic differences and environmental influences, the calibrated hydrophysiological parameters were acceptable. However, as shown in Table 4, several absolute correlation coefficients among calibrated parameters were over 0.8, which indicates parameter uncertainty might exist due to the equifinality of different parameters for the simulations of g_s and SWC. Thus, more observations regarding leaf water potential, root water potential, xylem ABA concentration, root aquaporin activity, root anatomy, and root-soil air gap are needed, which would contribute to further reducing the uncertainty of model parameters.

4.3. Simulation scenario I: Effects of different soil-root hydraulic resistance variations on water transport

As shown in Figs. 6–8, the individual and interactive effects of three soil-root hydraulic resistance variations on the hydraulic traits of plants are simulated with the calibrated hydrophysiological parameters (Table 3). Systematic analysis of the root-soil hydraulic resistance variation effects helps elucidate the interactions between stomatal regulation and soil-root hydraulic resistance variations in Section 4.4.

4.3.1. AQP scenario

The absolute water potentials of the root collar ($|h_{collar}|$) and leaf ($|h_l|$) (Fig. 6f and b) in the AQP scenario are lower than those in the Constant scenario when the absolute average soil water potential ($|h_s, \text{mean}|$) is below 0.45 MPa during the initial simulation period (before the 30th days) (Fig. 6d). The corresponding RWU is slightly enhanced (Fig. 7d), maintaining $0.012\text{--}0.013 \text{ cm}^3 \text{ cm}^{-3} \text{ d}^{-1}$ at a soil depth of 20–60 cm from the 8th to the 24th day. These positive hydraulic responses are attributed to the increase in K_{rr} in the AQP scenario (Fig. 7c), which alleviates the potential gradient and increases the flux transmission from the rhizosphere to the leaves in mild drought conditions. Sade et al. (2009), Chaumont and Tyerman (2014), and Gambetta et al. (2017) verified these promotion effects of aquaporin responses on the water permeability of root cells, root hydraulic conductivity, and plant transpiration. However, similar to the simulation case of “Aquaporin” in Couvreur et al. (2014b), these competitive AQP effects gradually disappeared in the later simulation stage. This is mainly because the soil hydraulic resistance gradually replaces the root radial hydraulic resistance as the primary restriction in the soil-root water path as the soil dries out, thereby limiting the promotion effect of K_{rr} improvement on soil-root water transport (Newman, 1969; Liu et al., 2015; Carminati and Javaux, 2020).

4.3.2. APD scenario

The value of $|h_{0, \text{mean}}|$ approach 1.8 MPa after the 30th day in the APD scenario (Fig. 6e), which is lower than that in the Constant scenario

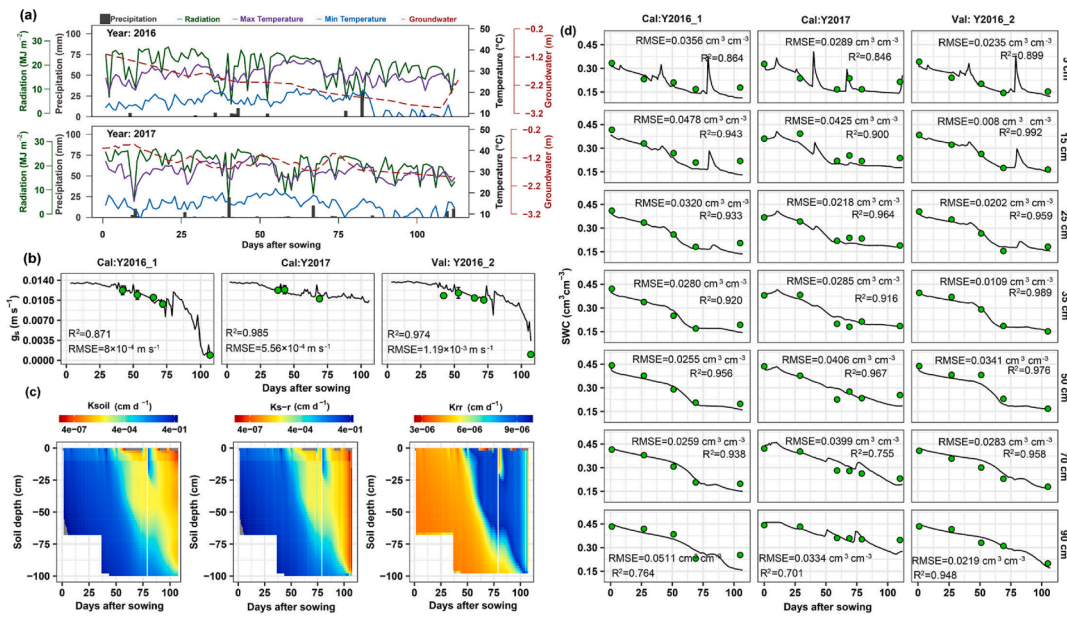


Fig. 5. Calibration and validation of the SR-HRV model using the field experiments conducted in the Y2016_1 and Y2016_2 and Y2017 plots. The left upper panel shows (a) daily meteorological data, including precipitation, global radiation, downward shortwave radiation, maximum temperature, minimum temperature, and groundwater table in 2016 and 2017. The left middle panel shows (b) the daily stomatal conductance (g_s) of sunflowers. The left lower panel shows the dynamic hydraulic conductivity at different depths during the whole simulation for the hydraulic conductivity of bulk soil (K_{soil}), soil-root interface (K_{s-r}), and root radial tissues (K_{rr}) in the validation (Y2016_2). The comparisons of the calibrated and measured soil water contents (SWCs) at different soil depths are shown in the right panel (d).

Table 4

Correlation coefficient matrix of calibrated parameters of the SR-HRV model for sunflower.

	L_{stem}	K_{rr}^*	$h_{sr,2}$	$h_{root,4}$	R	b	α	β	δ
L_{stem}	1.00	−0.04	−0.40	0.38	0.14	0.48	−0.32	−0.69	−0.25
K_{rr}^*		1.00	0.29	−0.30	−0.38	−0.37	0.48	0.07	0.04
$h_{sr,2}$			1.00	−0.84	−0.23	−0.86	0.45	0.88	−0.14
$h_{root,4}$				1.00	0.13	0.85	−0.51	−0.80	0.09
R					1.00	0.09	0.11	−0.21	−0.12
b						1.00	−0.79	−0.81	−0.16
α							1.00	0.34	0.33
β								1.00	−0.06
δ									1.00

(2.6 MPa). However, $|h_{collar}|$ and $|h_l|$ increase faster than in the Constant scenario (Fig. 6b and f), which are higher than 2.6 MPa on the 50th day. These water potential responses indicate the inhibition of hydraulic transfer efficiency between the xylem and root surface due to the APD effects (Fig. 7c), which restricts the water potential decrease in the root surface, thereby impeding the root water absorption. As shown in Fig. 6a, T_a is slightly lower than it is in the Constant scenario after the 40th day. Moreover, from a modeling perspective (according to Equation (13)), the inefficient water potential decrease in the root surface alleviates the increase in $|h_{plant}|$ and C_{ABA} . Therefore, C_{ABA} in the APD scenario is slightly lower than that in the Constant scenario (Fig. 6c). Interestingly, in a partial root-zone drying experiment for potato (*Solanum tuberosum* L.), Puertolas et al. (2015) found that the root ABA concentration accumulation of dry compartments in vertical heterogeneous conditions (upper dry) was lower than that in horizontal heterogeneous conditions. According to recent developments in the field of ABA biosynthesis and transport in a plant (McAdam et al., 2016; Kurumori et al., 2018; McAdam and Brodribb, 2018), the ABA is mainly generated in leaves and supplied to the roots via the phloem, the lower ABA accumulation of the older roots in the upper layers implies that root aging or tissue damage (e.g., APD) might inhibit the ABA synthesis in leaf and limits the ABA concentration in the xylem.

4.3.3. CONTACT scenario

Different from the water potential status in the APD scenario, $|h_0|$, $|h_{collar}|$, and $|h_l|$ constantly increased to 2.5–2.8 MPa after the 50th day in the CONTACT scenario (Fig. 6b, e and f), greater than those in the Constant scenario. However, the increase in $|h_{s,mean}|$ gradually approaches 1.0 MPa (Fig. 6d), which is lower than that in the Constant scenario (1.5 MPa). These phenomena are caused by hydraulic conductivity reduction in the rhizosphere (Fig. 7b), which alleviates water potential decrease in the bulk soil but accelerates the decrease in the root surface and xylem, thereby slightly increasing the C_{ABA} level (Fig. 6c). With the combined effects of h_l and C_{ABA} , T_a further decreases below the Constant scenario from the 30th to the 60th day (Fig. 6a). Therefore, we conjecture that the stomatal aperture, rather than merely relying on the hydraulic signal stimulation, is also closely regulated by the ABA or other chemical signals when the air gap gradually appears. This positive stimulation of chemical signal probably can explain the experimental results of Carminati et al. (2013) and Koebernick et al. (2018), who observed a proactive decrease in transpiration before the occurrence of CONTACT.

4.3.4. A + A + C scenario

With the interactive effects of three soil-root hydraulic resistance variations (A + A + C), the hydraulic responses of plants vary over time.

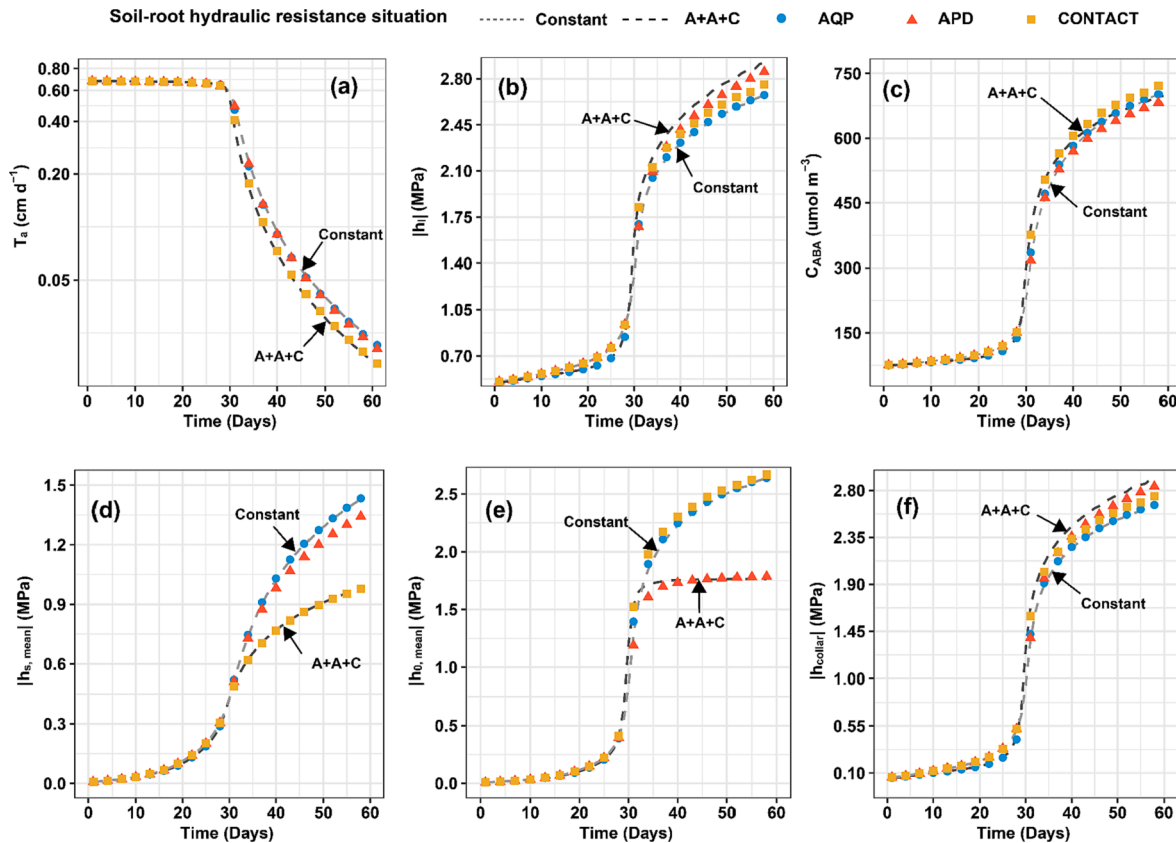


Fig. 6. Daily (a) transpiration (T_a), (b) absolute leaf water potential (h_l), (c) xylem ABA concentration (C_{ABA}), and absolute average water potentials of the 0–100 cm profile in (d) soil ($|h_{s,mean}|$), (e) root surface ($|h_{o,mean}|$), and (f) root collar ($|h_{collar}|$) with respect to the individual effects (solid line) of root aquaporins promotion (AQP), root apoplastic path damage (APD), root-soil contact loosening (CONTACT), and their interactive effects (A + A + C, dash-and-dotted line). The responses to the Constant hydraulic resistance of the soil-root system are also presented (dashed line).

Initially, the water potentials of the leaf and root collar are similar to those in the AQP scenario before the 30th day (Fig. 6b and f). The RWU is also enhanced at this stage (Fig. 7d). At the later simulation stage, the decreases in $|h_{s,mean}|$ (Fig. 6d) and $|h_{o,mean}|$ (Fig. 6e) are restricted by the effects of CONTACT and APD, respectively. While the variations of T_a (Fig. 6a) and RWU (Fig. 7d) in the A + A + C are similar to the CONTACT scenario rather than the APD scenario. Meanwhile, as shown in Fig. 8a, the relation between T_a and $|h_{s,mean}|$ in the A + A + C scenario is close to the CONTACT scenario. These results indicate that CONTACT restricts the response of water flux, including T_a (equivalent to stomatal conductance) and RWU, to severe soil dehydration. From a modeling perspective, this is because, although APD prevents the water potential decrease at the root surface, it is also low enough (-0.9 MPa ~ -1.8 MPa) to trigger the CONTACT (< -0.05 MPa) and causes a matric flux potential drop-down at the soil-root interface. However, C_{ABA} and $|h_l|$ in the A + A + C scenario are similar to those in the APD scenario (Fig. 6b and c), and as shown in Fig. 8b, the relation between T_a and $|h_l|$ in the A + A + C scenario is closer to the APD scenario. These results indicate that APD controls the drought signal responses and their regulation on T_a (equivalent to stomatal conductance).

Therefore, we conjecture that the hydraulic traits in the A + A + C scenario are determined by the unique and targeted effects of individual soil-root hydraulic resistance variation rather than their simple accumulation. Significantly, the occurrences and dynamic effects of CONTACT and APD should be particularly considered when plants suffer from severe drought stress, because they regulate the water flux and drought signal responses, respectively. Meanwhile, it is worth noting that CONTACT and APD are two dynamic inter-enforcing processes. As stated by Carminati et al. (2013), as a result of root apoplastic path damage (APD) and root shrink, air gap formation (CONTACT) increases

the hydraulic resistance in the rhizosphere, which further accelerates APD. Moreover, owing to the unique effects of CONTACT and APD on soil and root water potential (Fig. 6d and e), the observations of soil and root water potential might be the appropriate variables to parameterize these two processes when the plant is in a drought situation. Furthermore, complex hydrophysiological factors, e.g., soil texture and density, root hair activity, and mucilage secretion, are involved in the APD and CONTACT processes (Carminati et al., 2016; Koebernick et al., 2018). How to macroscopically quantify these effects and eliminate the scale effects between experimental observations and parameter determinations remains an open question.

4.4. Simulation scenario II: Effect of stomatal regulations on the variation in root-soil hydraulic resistance

Taking the situation simulated with calibrated stomatal parameters as a benchmark (SEN_10), the deviations of T_a and relations between $|h_l|$ and $|h_{s,mean}|$ (Figs. 9 and 10) in different soil-root hydraulic resistance variation scenarios and stomatal sensitivity situations are compared in this section.

4.4.1. Stomatal sensitivity

In the SEN_30 situations, stomata are more sensitive to drought signals than they are in the SEN_10 situations. Plants maintain high transpiration of $0.60\text{--}0.65$ cm d $^{-1}$ for a shorter time (0–15 days), while the lower water consumption alleviates the decrease in T_a at the later stage (Fig. 9a). Moreover, as shown in Fig. 10, the sensitive stomatal responses, in the SEN_30 situations, prevent the increases in $|h_l|$ (< 1.6 MPa) and C_{ABA} (< 400 $\mu\text{mol m}^{-3}$) as $|h_{s,mean}|$ increase. These results imply a cautious stomatal strategy (more isohydric behavior) of the

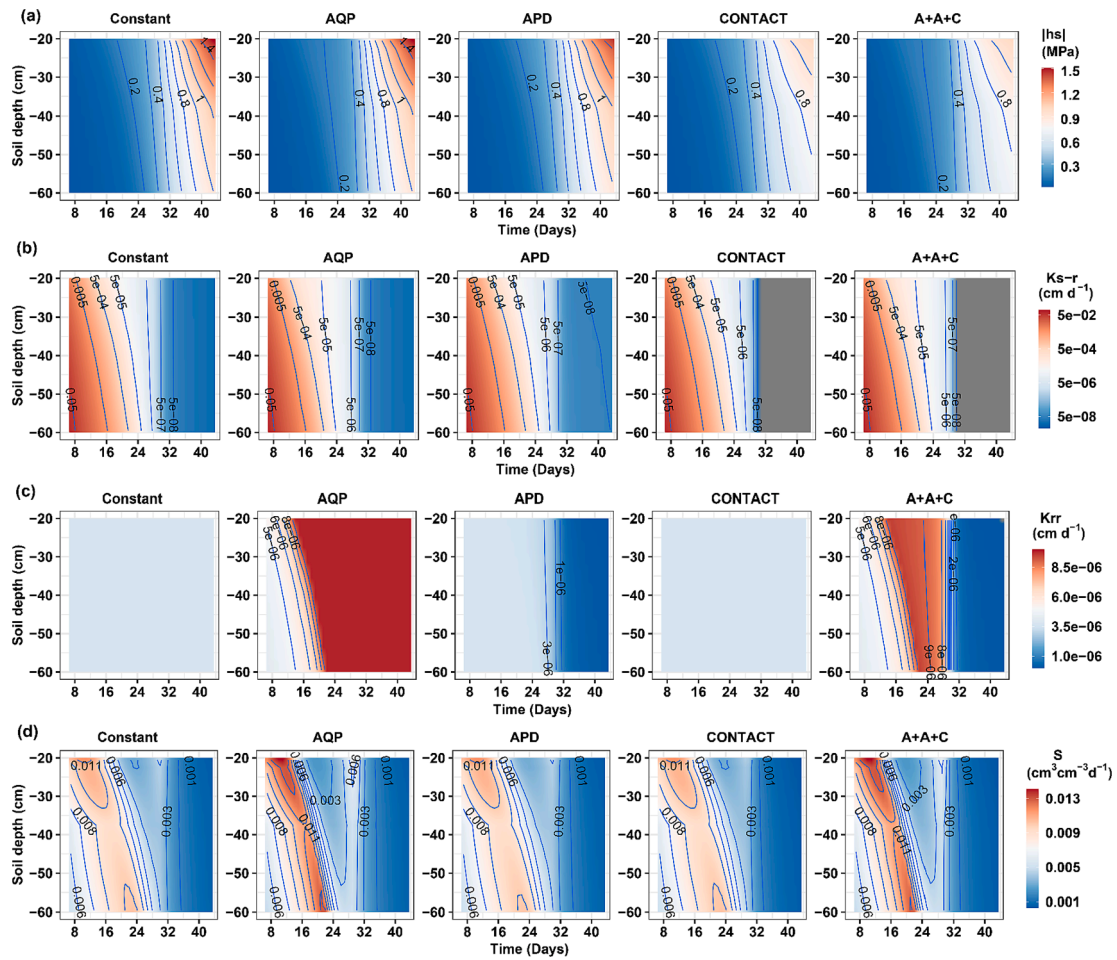


Fig. 7. Dynamic distributions of (a) absolute soil water potential ($|h_s|$), (b) hydraulic conductivity in the rhizosphere (K_{s-r}), (c) root radial conductivity (K_{rr}), and (d) root water uptake (S) at soil depths from -20 cm to -60 cm from the 8th to the 40th day with the individual effects of root aquaporins promotion (AQP), root apoplastic path damage (APD), root-soil contact loosening (CONTACT), and their interactive effects (A + A + C) on the soil-root hydraulic conductivity. The distributions with constant hydraulic resistance of the soil-root system are also presented.

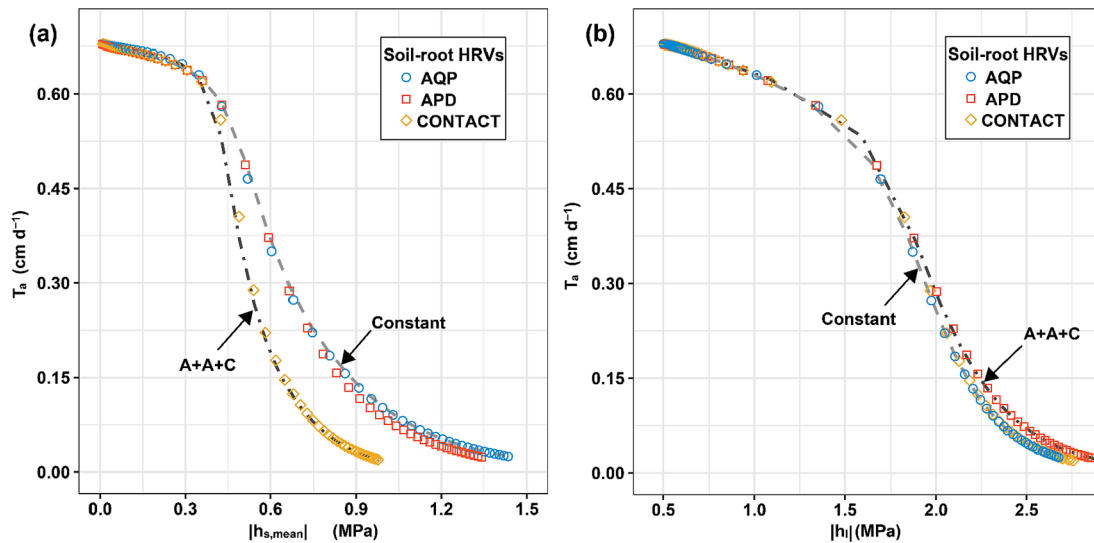


Fig. 8. (a) Relations between transpiration (T_a) and absolute average water potentials of the 0–100 cm profile in soil ($|h_{s, \text{mean}}|$); (b) relations between T_a and absolute leaf water potential ($|h_l|$). These relations with respect to the individual effects (solid line) of root aquaporins promotion (AQP), root apoplastic path damage (APD), root-soil contact loosening (CONTACT), and their interactive effects (A + A + C, dash-and-dotted line). The responses to the Constant hydraulic resistance of the soil-root system are also presented (dashed line).

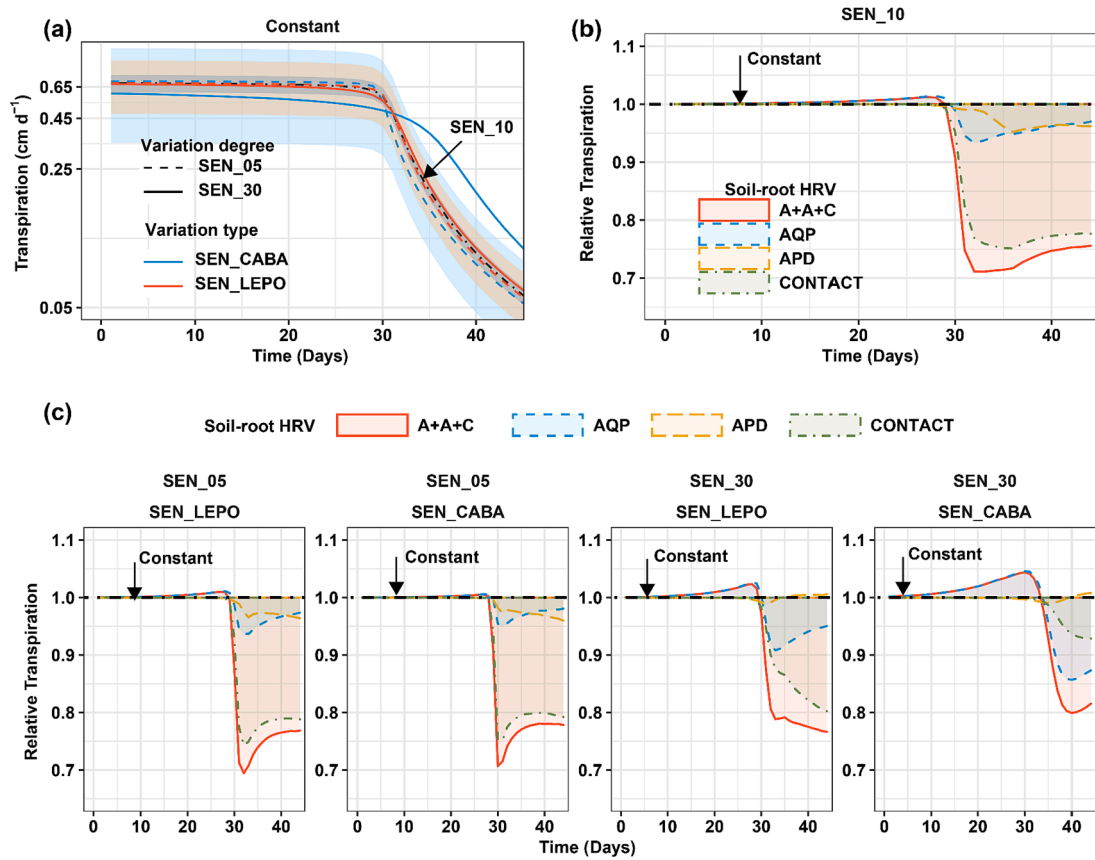


Fig. 9. (a) Daily transpiration comparisons with different stomatal sensitivities in the Constant scenario. The stomatal sensitivities of SEN_05 (dotted lines), SEN_10 (dotted-dashed lines), and SEN_30 (solid lines) to the xylem ABA concentration (SEN_CABA, light blue lines) and leaf water potential (SEN_LEPO, red lines) are regulated by the parameters β ($\beta = -0.002295, -0.00459, \text{ and } -0.01377$) and δ ($\delta = -0.11, -0.22, \text{ and } -0.66$) in Equation (12). The black dotted-dashed line represents transpiration with calibrated stomatal properties (benchmark status), and we extended this to 0.7–1.3 and 0.5–1.5 times the benchmark, denoted by the light red and blue shading, respectively. The relative transpiration of soil-root hydraulic resistance variations (hydraulic resistance variations, including the root aquaporins promotion (AQP), root apoplastic path damage (APD), root-soil contact loosening (CONTACT), and their interactive effects (A + A + C)) compared with Constant scenario in the (b) SEN_10 situation and (c) other stomatal sensitivity situations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

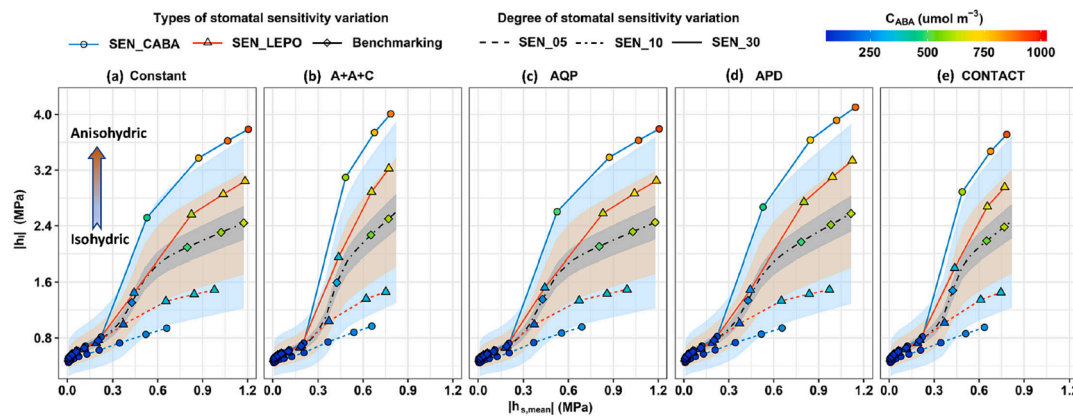


Fig. 10. Relations between absolute leaf water potential ($|h_l|$) and absolute average water potentials of the 0–100 cm profile in soil ($|h_{s,mean}|$) for plants with different stomatal sensitivities to C_{ABA} (SEN_CABA, light blue lines) and the leaf water potential (SEN_LEPO, red lines), which are regulated by the parameters β and δ in Equation (12), respectively. These relations are presented when daily transpiration is greater than 0.01 cm d^{-1} . The situations of SEN_05 (dotted lines) and SEN_30 (solid lines) with halved and doubled β and δ values are compared to the calibrated situation ($\beta = -0.00459$ and $\delta = -0.22$) (SEN_10, dotted-and-dashed lines). The black dotted-dashed line represents transpiration with calibrated stomatal properties (benchmark status), and we extend it to 0.7–1.3 and 0.5–1.5 times the benchmark, denoted with the light red and blue shading, respectively. Besides the response to (a) Constant soil-root hydraulic conductivity, the transpiration responses to the soil-root hydraulic resistance variations including the (c) root aquaporins promotion (AQP), (d) root apoplastic path damage (APD), (e) root-soil contact loosening (CONTACT), and (b) their interactive effects (A + A + C), are simulated. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

plants to adapt to long-term drought stress (Martin-StPaul et al., 2017). However, when stomata are less sensitive to the drought signals in the SEN_05 situations (more anisohydric behavior), the opposite traits of T_a are presented as time goes by, plants maintain high transpiration of $0.65\text{--}0.7\text{ cm d}^{-1}$ for a longer time (0–30 days), the greater water consumption accelerates the decrease in transpiration in the later stage (Fig. 9a). The corresponding $|h_i|$ (increase to $2.5\text{--}3.8\text{ MPa}$) and C_{ABA} (over $800\text{ }\mu\text{mol m}^{-3}$) rapidly increase as $|h_{s,\text{mean}}|$ increase in SEN_05 situations (Fig. 10a).

4.4.2. Performances of AQP, APD, and CONTACT

Distinct hydraulic responses are presented in different soil-root hydraulic resistance variation scenarios and stomatal sensitivity situations. Compared with the Constant scenario in the SEN_10 situation (Fig. 9b), the relative transpiration in the AQP scenario is greater than 1 and gradually increases before the 27th day, while after that, it begins to decrease and decrease below 0.95 from the 31st to 36th day. Interestingly, the variation ranges of relative transpiration in the AQP scenario are reduced and amplified in the SEN_05 and SEN_30 situations (Fig. 9c), respectively. These simulation phenomena imply that the AQP effects are constrained under insensitive stomatal situations when plants with more anisohydric behavior. From the modeling perspective, one possible reason is that the AQP mainly acts at the initial simulation stage when the soil is relatively wet (as described in Section 4.3). Only a few hydraulic ($|h_i| < 1.3\text{ MPa}$) and ABA ($C_{ABA} < 200\text{ }\mu\text{mol m}^{-3}$) signals are produced from the AQP effects to regulate the stomatal aperture during this period. These disturbances are diluted for plants with insensitive stomatal responses because of the relatively high drought signal level in this situation, as shown in the SEN_05 situation (Fig. 10c). However, paradoxically, the root aquaporin activity of anisohydric plants seems to be more active than isohydric plants, such as Vandeleur et al. (2009) found that the permeability of root cortex cells of Chardonnay (an anisohydric cultivar of grapevine) is higher than Grenache (an isohydric cultivar of grapevine) under drought stress due to the upregulation of PIPs (plasma membrane intrinsic proteins). This phenomenon implies complex hydrophysiological responses of plants to drought, it is not enough to describe the plant hydraulic behavior merely at a local organ scale, more attention should be paid to the interactions between different organs and hydraulic behaviors in soil-plant water transport modeling.

In the APD and CONTACT scenarios, the relative transpiration is approximately 1.0 before the 30th day in the SEN_10 situation (Fig. 9b), but after then, it rapidly decreases to 0.95 and 0.7 on the 33rd–35th days. Similar variation ranges of relative transpiration are presented in the SEN_05 situation (Fig. 9c). However, in the SEN_30 situation (Fig. 9c), the variation ranges of relative transpiration are reduced, which indicates that the sensitive stomatal response protects plant transpiration from the influences of APD and CONTACT, especially APD. The reason is that the sensitive stomatal regulation restricts leaf water loss by immediate closure of stomata, thereby preventing the water potential from further decreasing in the soil-plant system, as shown in Fig. 10d and e. Similarly, Martin-StPaul et al. (2017), Rodriguez-Dominguez and Brodribb (2020), and Carminati et al. (2020) successively demonstrated the importance of stomatal closure to prevent xylem embolism and alleviate the formation of a large gradient in water potential around the roots. Moreover, due to the greater regulation capacity of CONTACT with respect to transpiration response to soil dehydration in the A + A + C scenario as explained in Section 4.3, similar relative transpiration responses are obtained in the CONTACT and A + A + C scenarios in the SEN_10 and SEN_05 situations (Fig. 9b and c). Furthermore, as shown in Fig. 10b and e, as $|h_{s,\text{mean}}|$ increases in the CONTACT and A + A + C scenarios, the $|h_i|$ and C_{ABA} rapidly increase, which would stimulate the stomatal closure and implies that CONTACT limits soil water availability of plants. However, as a result of the effects of APD on drought signal responses (as explained in Section 4.3), as $|h_{s,\text{mean}}|$ increases, the increases in $|h_i|$ in A + A + C scenarios are slightly greater than those in

CONTACT scenarios.

4.4.3. Regulations of hydraulic and chemical signals

Compared with the benchmarking situation, the deviations of T_a , $|h_i|$, and C_{ABA} in the SEN_CABA situations are greater than those in the SEN_LEPO situations (Figs. 9–10). These results are consistent with the sensitivity analysis of the ABA-related parameters in Section 4.1, indicating that the stomatal responses to ABA signals have significant impacts on water transport in the simulation. Similar simulation results were presented by Huber et al. (2015). An argument is put forward that only relying on the regulation of leaf water potential is insufficient to portray the stomatal behavior and hydraulic properties of plants with more anisohydric behaviors. An appropriate description of stomatal responses to chemical signals is necessary for these plants in water transport modeling. Furthermore, it is worth noting that the drought signals also participate in other hydraulic and physiological processes. For example, a lower plant water potential can induce hydraulic failure in the xylem (embolism) (McDowell et al., 2008; Martin-StPaul et al., 2017), and ABA concentration is also correlated with biomass assimilations of leaves and roots (Puertolas et al., 2015; Dinnyen, 2019), while these processes are not the focus of this paper. Moreover, a complex root-shoot signaling interaction is behind the ABA biosynthesis and accumulation in plants, while the xylem ABA concentration in the leaf was simply quantified by a function of the mean status of water potential in shoot and root in this study, which might induce the uncertainty in stomatal response to soil-root hydraulic variation, especially to APD. And due to the physiological and genetic differences among species, other chemical signals, such as the cytokinin and pH in leaf tissues, also participate in stomatal regulation (Davies et al., 2002; Chaves et al., 2003). The generation mechanism and regulation ability of these chemical solutes may affect the responses of g_s and h_i to drought (iso/anisohydric behavior). However, how to quantify the effects of these chemical solutes and whether these effects are necessary to consider in the stomatal model need more exploration in further studies.

5. Conclusions and outlooks

In this study, we proposed a soil-plant hydraulic model, SR-HRV, by improving the macroscopic physical RWU model of de Jong van Lier et al. (2008), de Jong van Lier et al. (2013) and explicitly describing the upper boundary with the Tardieu–Davies (1996), Tardieu–Davies (2015) stomatal conductance model. Compared with previous studies, the SR-HRV model innovatively bridges the gap with hydraulic and chemical signals to simulate the interactions between soil-root hydraulic resistance variations and stomatal regulation. The effects of three universal hydraulic resistance variations of root aquaporins promotion (AQP), root apoplastic path damage (APD), and root-soil contact loosening (CONTACT) were specifically characterized by piecewise linear functions. Based on the parameter sensitivity analysis and calibration, the dynamic interactions of soil-root hydraulic resistance variations and stomatal regulation on the hydraulic traits of plants were simulated in different scenarios. The following conclusions can be made:

- (1) The sensitivity of simulated g_s and SWC50 to stomatal regulation parameters, especially to the ABA signal correlated parameters, respond earlier and greater than the soil-root hydraulic resistance variation parameters. However, the sensitivity to APD and CONTACT parameters would rapidly increase with further soil dehydration.
- (2) Three soil-root hydraulic resistance variations distinctively affect the plant hydraulic traits in the A + A + C scenario. AQP improves root water absorption and affects transpiration when plants are in mild drought conditions. CONTACT and APD respectively regulate the water flux and drought signal responses when plants suffer severe drought stress.

- (3) The increase of stomatal sensitivity to drought signals induces early transpiration reduction but alleviates the transpiration decline for a longer time, which enhances the AQP effects and eliminates the effects of APD and CONTACT. Conversely, when stomatal sensitivity decrease to drought signals (especially to ABA signal), the opposite responses of soil-root hydraulic resistance variations effects are presented.

The above conclusions contribute to exploring the primary hydro-physiological response and balancing the simulation accuracy and model complexity in soil-plant water transport modeling. For plants with a more anisohydric behavior, the regulation of chemical signals on stomatal conductance should be considered. Moreover, it is essential to understand the stomatal sensitivity of plants, which determines the necessity to consider the effects of soil-root hydraulic resistance variations in water transport modeling. The primary influential factor of soil-root hydraulic resistance variation relies on the soil-plant water situation. Some hydrophysiological responses can be simplified in the simulation, such as the AQP effects in severe drought conditions and APD/CONTACT effects in mild drought conditions, while the interactive effects of APD and CONTACT should be particularly considered when plants suffer continuous drought stress.

Based on the calibrated hydrophysiological parameters, the SR-HRV model favorably simulated the stomatal conductance, soil water status, and soil-root hydraulic resistance variations at the field scale. However, some assumptions were made during the model development due to data limitations in the literature, such as the soil-root hydraulic resistance variation effects were quantified by the linear functions of hydraulic status in the soil-root system, the xylem ABA concentration was considered to be related to the average hydraulic status of the leaf and root. And to simplify the model, the root xylem resistance, the formation of xylem cavitation, and the hysteresis effects of stomatal response and soil-root hydraulic resistance variation were not considered in this study. These assumptions may have an impact on simulation results of soil-root hydraulic resistance, drought signal generation, and stomatal response, such as the quantification of xylem ABA concentration accounting for the synthetic effects of the shoot and root hydraulic states, which affects the interactions between stomatal regulation and soil-root hydraulic resistance variations. Thus, the rationality and potential effects of these assumptions need further clarification based on the detailed hydro-morpho-physiological observations in plant drought experiments. Moreover, exhaustive parameter calibration is beneficial for model application, for the parameters of stomatal regulation and xylem ABA concentration responses, it is recommended to employ convenient observations of stomatal conductance, xylem ABA concentration, and leaf water potential in calibrations. While for the parameter calibrations of soil-root hydraulic resistance variations, besides the essential measurements of soil water status and root water potential, the observations of root morphology, anatomy, and aquaporins activity are also advised. Furthermore, the interactions between stomatal regulation and soil-root hydraulic resistance variations should be tested under various soil types and atmospheric conditions. It is promising to couple the SR-HRV model with other physiological simulations, including carbon assimilation and biomass allocation, which provide a promising way to explore the influences of stomatal regulation and soil-root hydraulic resistance variations on crop growth, yield formation, and the water-carbon cycle.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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